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Problem Set Number: 4

Due Date: 3/3/25

Acknowledgement:

- checked work w/ Carmen + Rochelle

1. **Known:**

$t_g = 3\,mm$ | $\eta = a - bT_{si}$ | $G = 800\,W/m^3$ | $T_\infty = 300K = T_{sur}$

$k_g = 1.5\,W/(mK)$ | ↳ $a = 0.553$ | 5% reflected | $h = 10\,W/(m^2K)$

$t_a = 0.1\,mm$ | ↳ $b = 0.001K^{-1}$ | 10% glass absorbed | $h_{rad} = 5\,W/(m^2K)$

$k_a = 1\,W/(mK)$ | $R_c = 10^{-4}\,m^2K/W$ | 85% silicon absorbed

Find:

a) thermal resistance network : assumptions + values

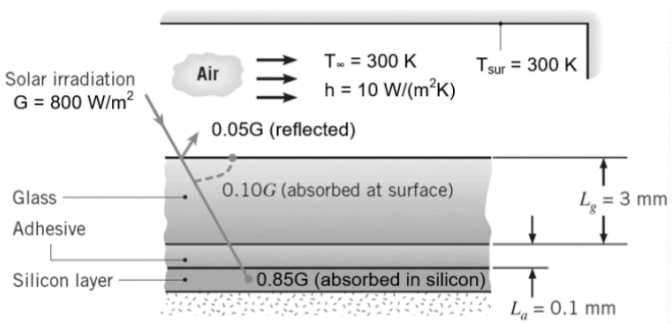
b) $T_{g,top} = ?$; $T_{si} = ?$

c) $\eta = ?$; $P = ?$

Assumptions:

- 1D heat transfer
- uniform properties

Schematic:



Analysis:

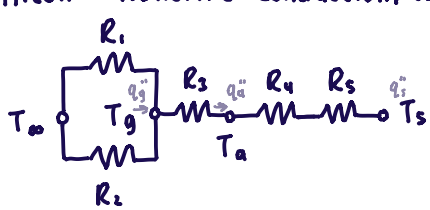
a) Air → Glass: radiation, convection

Adhesive → Glass : conduction

Silicon → Adhesive: conduction, contact

Assumptions:

- 1D heat transfer
- uniform properties



$$R_1 = \frac{1}{h_{rad}} = \frac{1}{5} = 0.2\,\frac{m^2K}{W}$$

$$R_2 = \frac{1}{h} = \frac{1}{10} = 0.1\,\frac{m^2K}{W}$$

$$R_3 = \frac{L_g}{k_g} = \frac{0.003}{1.5} = 0.002\,\frac{m^2K}{W}$$

$$R_4 = R_c = 10^{-4}\,m^2K/W$$

$$R_5 = \frac{L_a}{k_a} = \frac{0.0001}{1} = 0.0001\,\frac{m^2K}{W}$$

b) Glass:

absorbed by glass:

$q''_g = 0.1G = 80\,W/m^2$

energy balance:

$$q''_{si} + q''_g = \frac{\Delta T}{R_{th}} = \frac{T_{g,top} - 300}{(1/0.2 + 1/0.1)^{-1}} = q''$$

Silicon → Glass:

$$R_{tot} = 0.002 + 0.0001 + 10^{-4} = 0.0022\,\frac{m^2K}{W}$$

$$q'' = \frac{\Delta T}{R_{tot}} = \frac{T_{si} - T_{g,top}}{0.0022}$$

Silicon:

absorbed by silicon:

$1 - 0.553 + 0.001$

$$q''_{si} = 0.85G(1-\eta) = 0.85(800)(1-(0.553 - 0.001T_{si}))$$
$$= 303.96 + 0.68T_{si}$$

$$\frac{T_{si} - T_{g,top}}{0.0022} = \frac{T_{g,top} - 300}{0.067} = 303.96 + 0.68T_{si} + 80$$

$$T_{si} = \frac{1}{0.68}\left[\frac{T_{g,top} - 300}{0.067} - 383.96\right]$$

plugging in + plotting →

$T_{g,top} = 341.12\,K$

$T_{si} = 342.41\,K$

c) $\eta = 0.553 - 0.001T_{si}$

$= 0.553 - 0.001(342.41) = 0.21$

$q'' = \eta(0.85G)$

$= 0.21(0.85(800)) = 142.8\ W/m^2$

2. **Known:**

$D = 10$ mm | $L_{ins} = 200$mm | $T_{max} = 100°C$

$k = 80$ W/(mK) | L_o | $T_\infty = 20°C$

$T_w = 200°C$ | insulated tip | $h = 20$ W/(m²K)

Find:

a) Expression for T_o

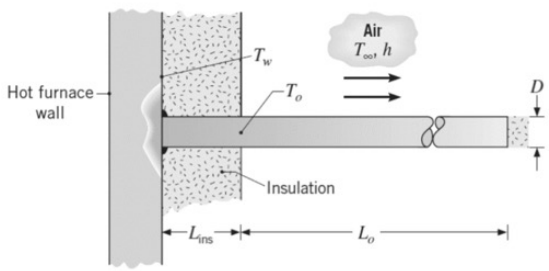
b) $L_o = 200$mm → $T_o = ?$
meet limit?

c) $L_o = ?$

Assumptions:

- 1D conductive HT
- steady state
- perfect insulator

Schematic:



Analysis:

a) $q_f = \sqrt{hPkA_c}\,\theta_b \tanh(mL_o)$

$$q'' = \frac{k(200 - T_o(0))}{L_{ins}} = 0$$

$T_o(0) = 200°C$

$$q_{cond} = kA_c \frac{T_w - T_o}{L_{ins}}$$

$q_f = q_{cond}$

$$\sqrt{hPkA_c}\,(T_o - T_\infty)\tanh(mL_o) = kA_c \frac{T_w - T_o}{L_{ins}}$$

$$\sqrt{hPkA_c}\,T_o \tanh(mL_o) - \sqrt{hPkA_c}\,T_\infty \tanh(mL_o) = \frac{kA_c}{L_{ins}} T_w - \frac{kA_c}{L_{ins}} T_o$$

$$T_o = \frac{(kA_c/L_{ins})T_w + \sqrt{hPkA_c}\,T_\infty \tanh(mL_o)}{\sqrt{hPkA_c}\tanh(mL_o) + (kA_c/L_{ins})}$$

$R = D/2 = 5$mm

$P = 2\pi R$ $\qquad A_c = \pi R^2$

$$m = \sqrt{\frac{hP}{kA_c}} = \sqrt{\frac{20(2)}{80(0.005)}} = 10$$

$$T_o = \frac{200 + 40\tanh(10L_o)}{2\tanh(10L_o) + 1}$$

b) $T_o(L_o = 0.2) = 81.47°$

↳ within limit

c) set $T_o = 100$ and plot

$L_o = 73.32$mm

3. **Known:**

- square pin fin
- $k = 400\ W/mK$
- $S = 10\ mm$
- $h = 100\ W/m^2K$
- $\eta_f = 0.6$
- convective tip

Find:

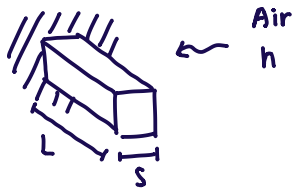
a) $L = ?$

b) $R_f = ?$, $\varepsilon_f = ?$

Assumptions:

- uniform cross-section area

Schematic:



Analysis:

a) $$\eta_f = \frac{\tanh(mL_c)}{mL_c}$$

$$m = \sqrt{\frac{hP}{kA_c}} = \sqrt{\frac{100(4(0.01))}{400(0.01^2)}} = 10\ m^{-1}$$

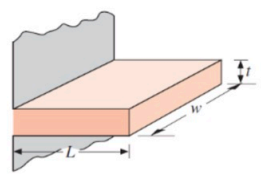
$$0.6 = \frac{\tanh(10L_c)}{10L_c}$$

$$L_c = 151\ mm$$

$$L = L_c - t/2$$

$$= 0.151 - 0.01/2 = 146\ mm$$

Rectangular[a]	
$A_f = 2wL_c$	
$L_c = L + (t/2)$	
$A_p = tL$	



$$\eta_f = \frac{\tanh mL_c}{mL_c}$$

b) $$R_f = \frac{1}{\eta_f h A_f}$$

$$= \frac{1}{0.6(100)(4(0.146)(0.01) + 0.01^2))} = 2.81\ \frac{K}{W}$$

$$\varepsilon_f = \sqrt{\frac{kP}{hA_c}}\tanh(mL_c) = \sqrt{\frac{400(4)}{100(0.01)}}\tanh(mL_c) = 36.28$$